

Introduction

To begin this task, I downloaded and installed the WEKA datamining software. I launched the WEKA GUI Chooser, opened the 'Explorer' interface, and navigated to the 'Preprocess' tab. From there, I opened the folder containing the preinstalled datasets and loaded my first chosen dataset: iris.arff. I then navigated to the 'Classify' tab to begin testing the six required algorithms.

Iris Section

Naïve Bayes

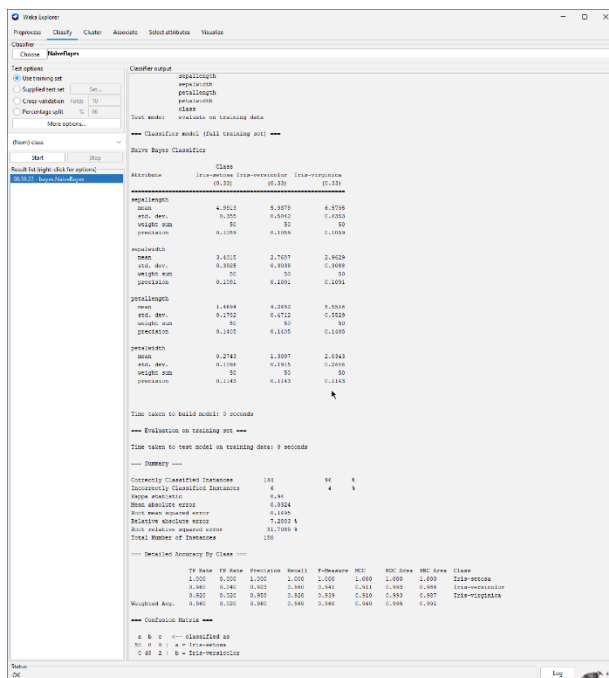


Figure 1: Naïve Bayes, Use Training Set

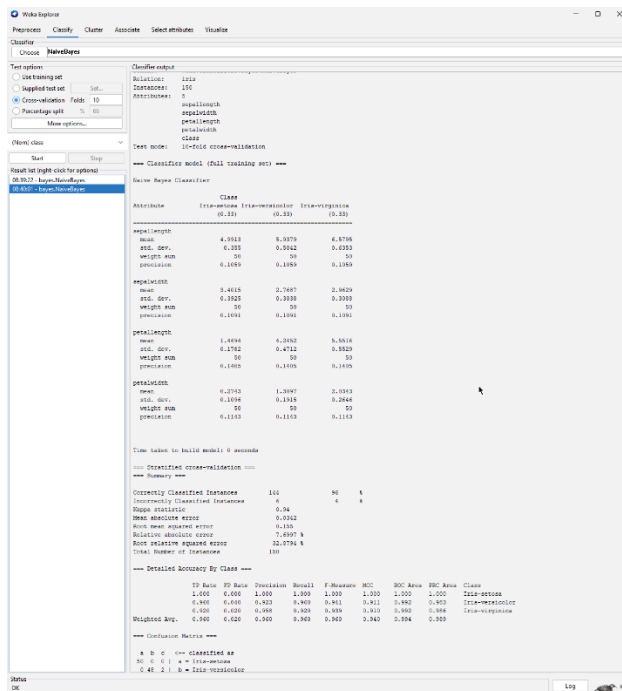


Figure 2: Naïve Bayes, Cross-validation (10 Folds)

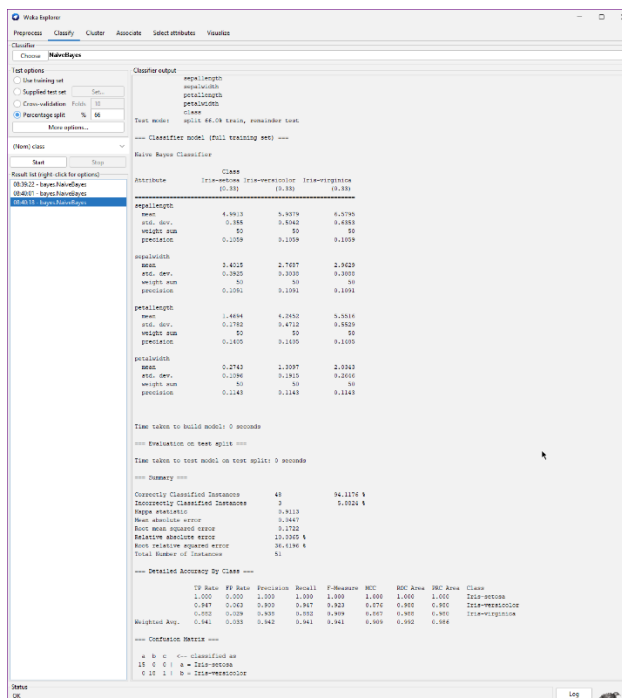


Figure 3: Naïve Bayes, Percentage Split (80%)

Logistic Regression

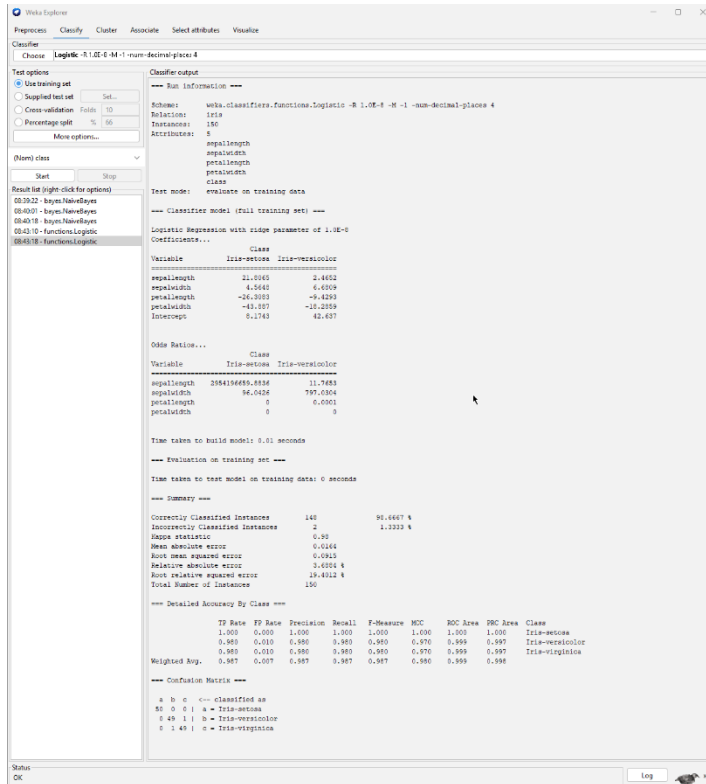


Figure 4: Logistic Regression, Use Training Set

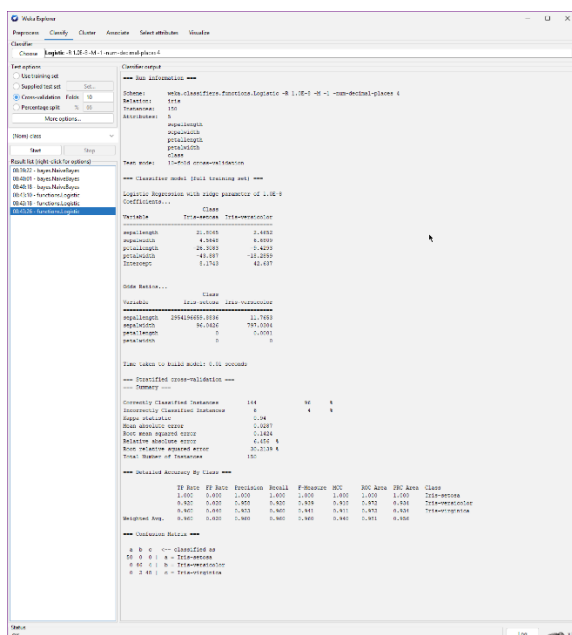


Figure 5: Logistic Regression, Cross-validation (10 Folds)

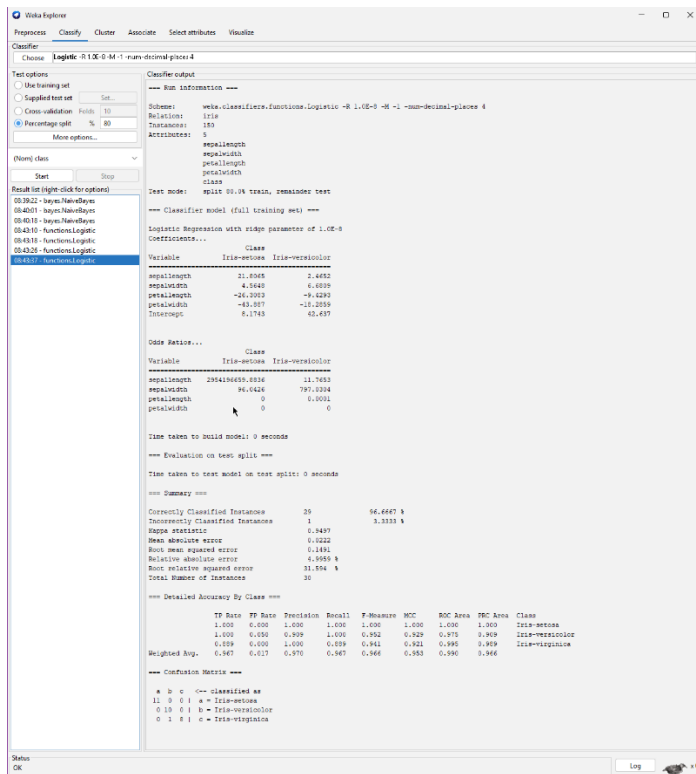


Figure 6: Logistic Regression, Percentage Split (80%)

SMO

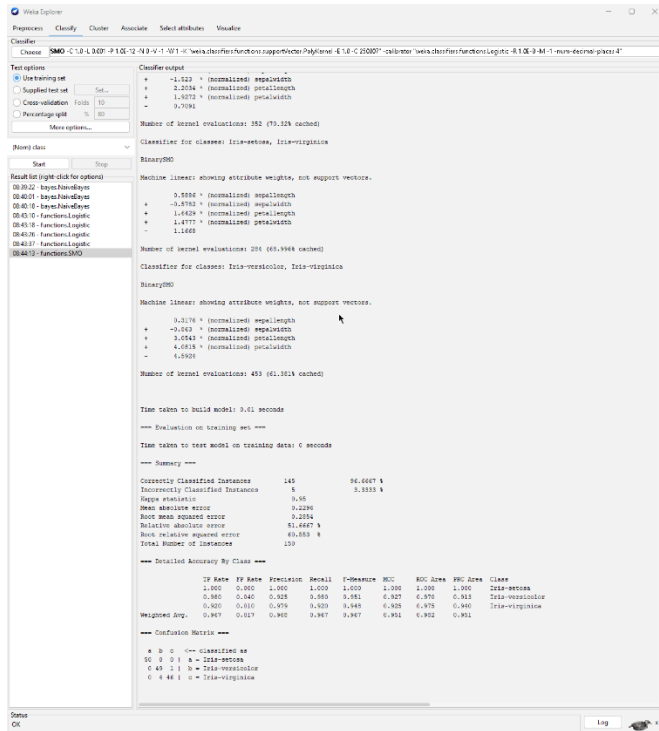


Figure 7: SMO, Use Training Set

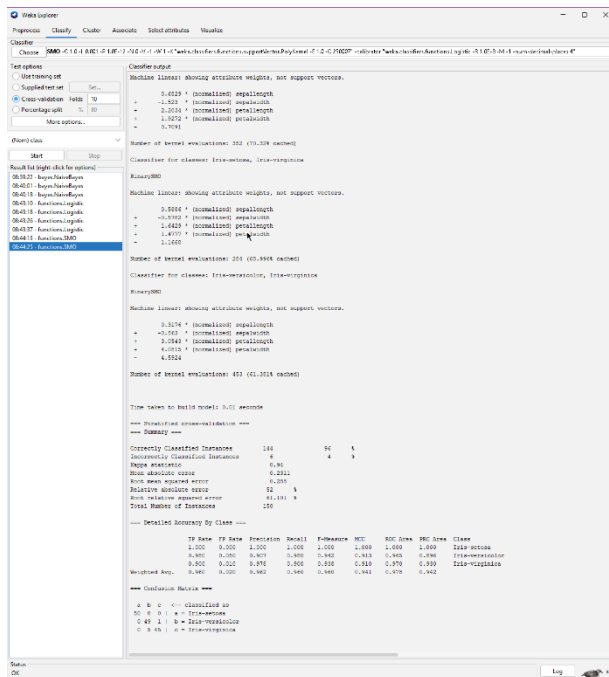


Figure 8: SMO, Cross-validation (10 Folds)

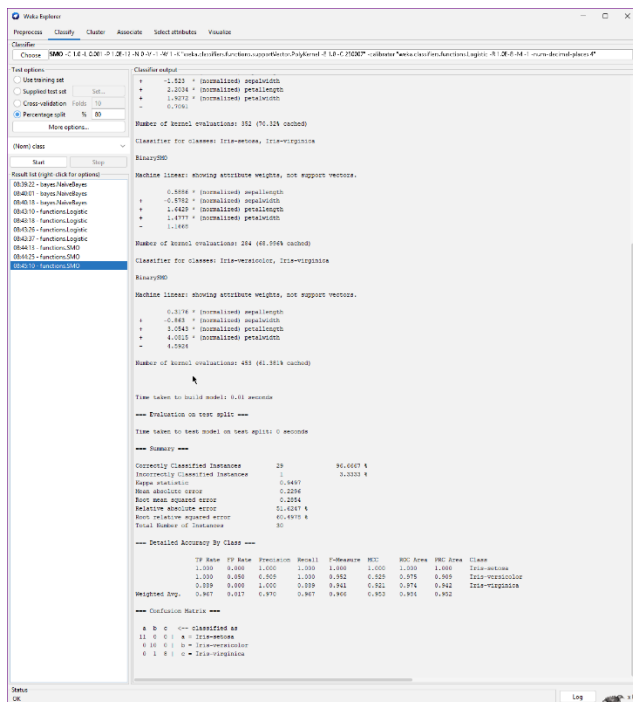


Figure 9: SMO, Percentage Split (80%)

K-Nearest Neighbor

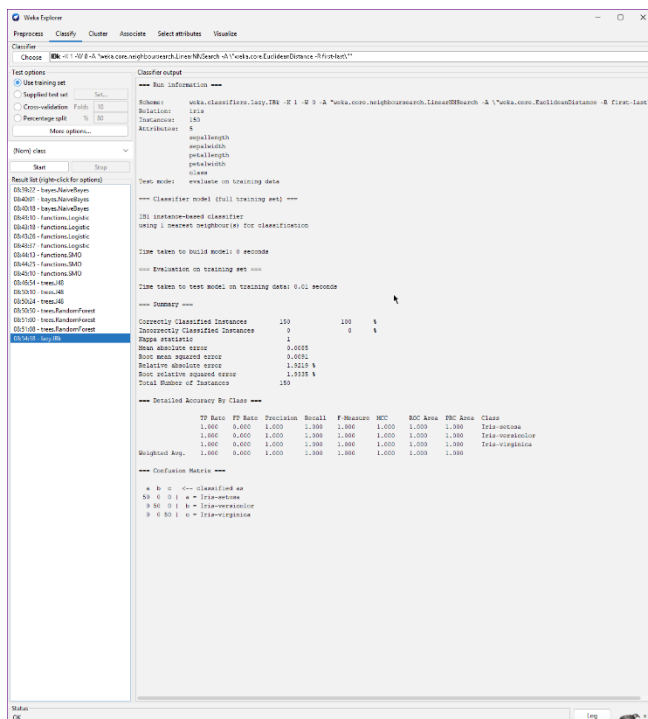
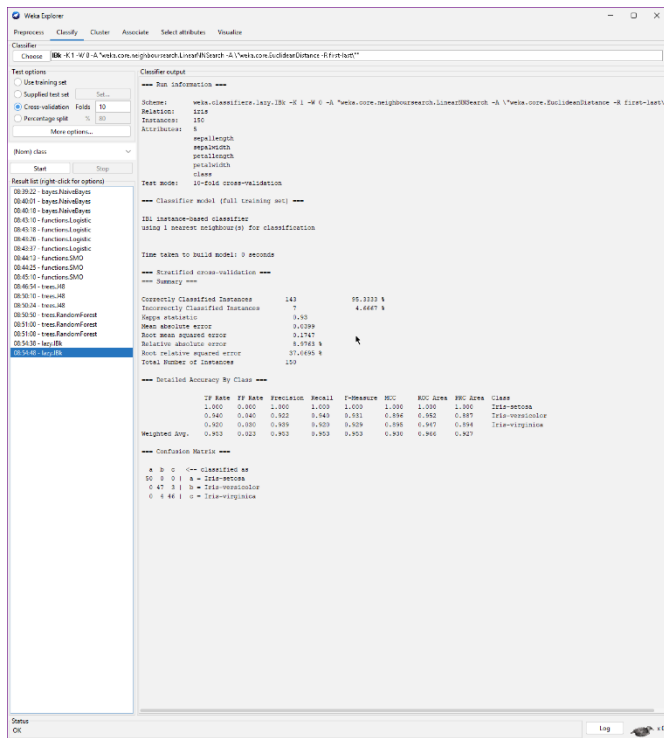


Figure 10: K-Nearest Neighbor, Use Training Set



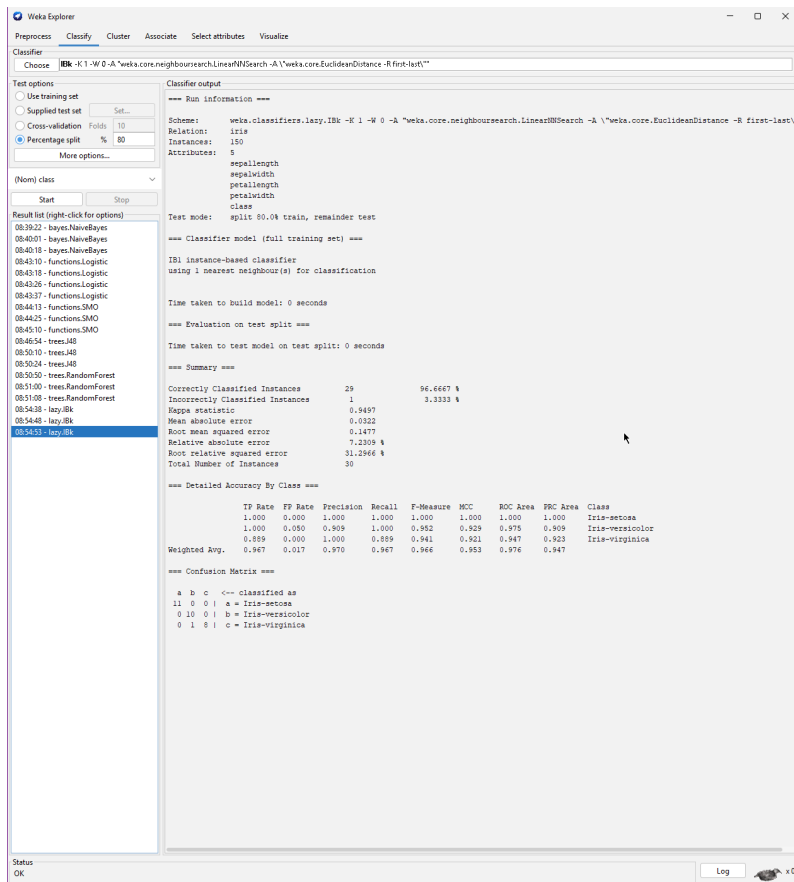


Figure 12: K-Nearest Neighbor, Percentage Split (80%)

J48

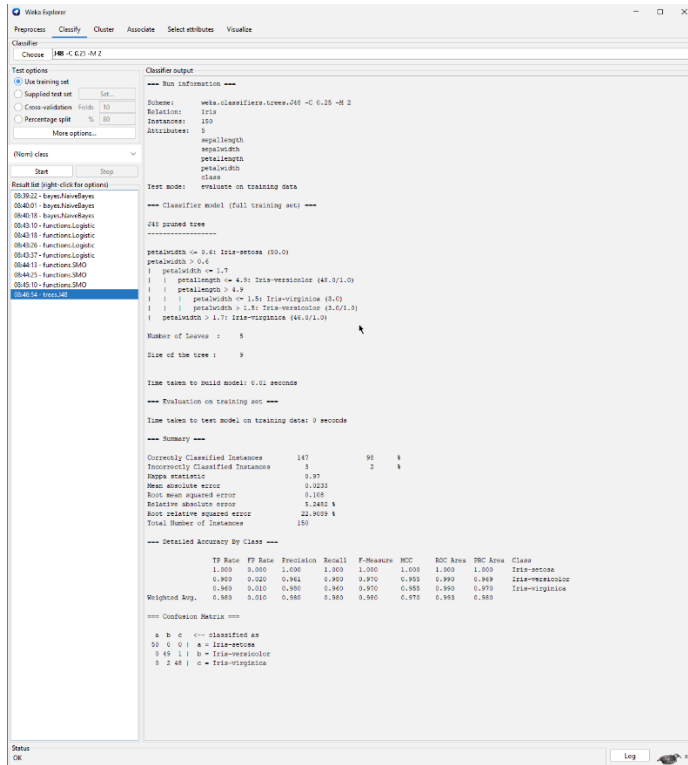


Figure 13: J48, Use Training Set

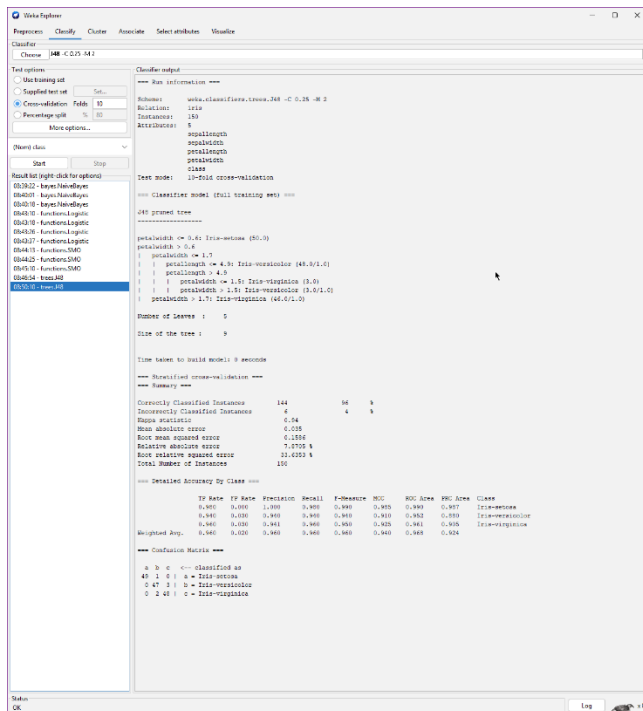


Figure 14: J48, Cross-validation (10 Folds)

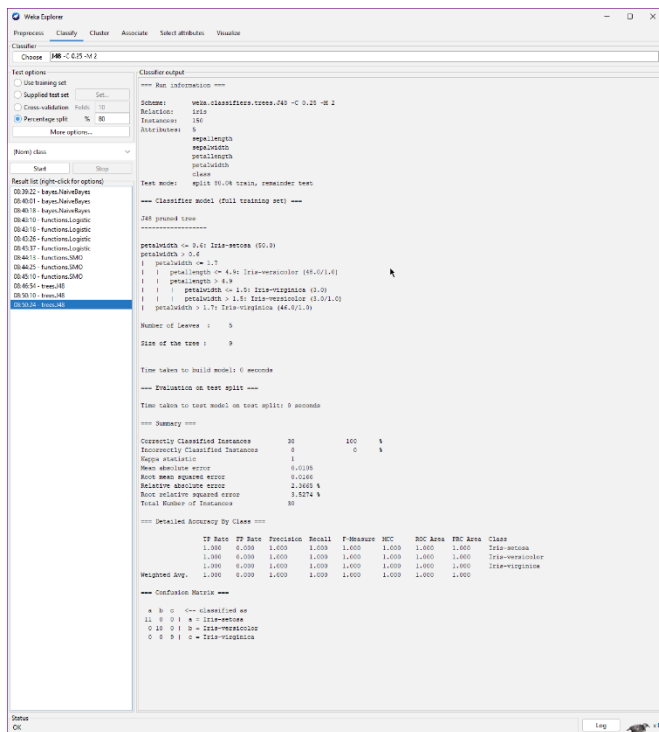


Figure 15: J48, Percentage Split (80%)

Random Forest

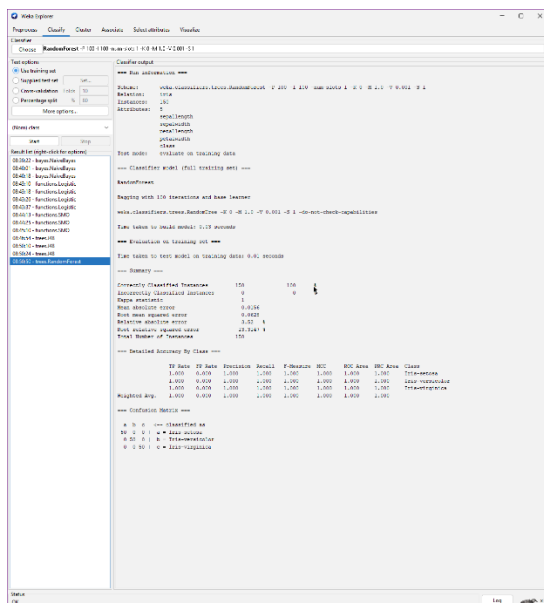


Figure 16: Random Forest, Use Training Set

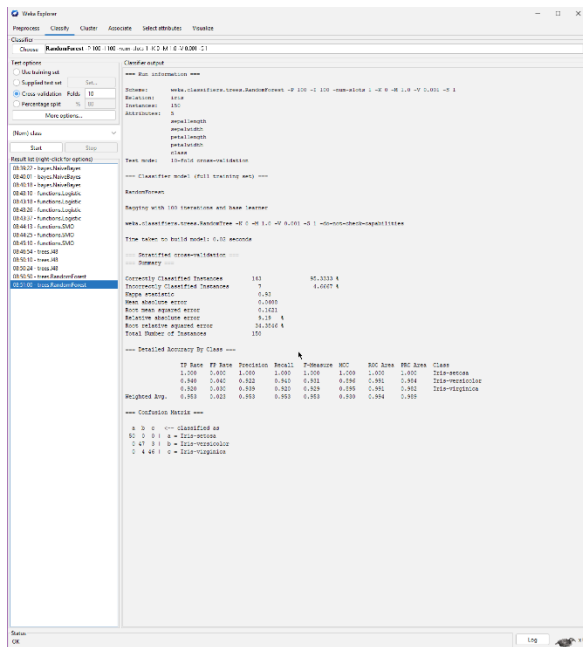


Figure 17: Random Forest, Cross-validation (10 Folds)

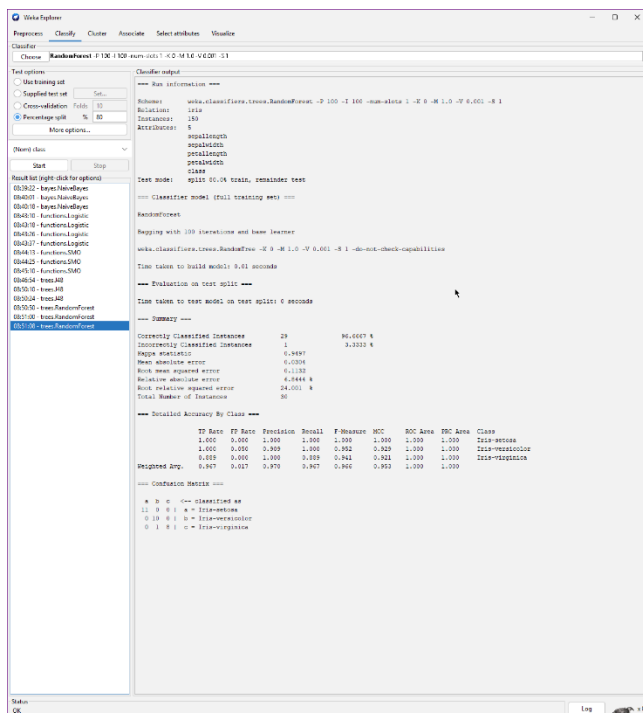


Figure 18: Random Forest, Percentage Split (80%)

Diabetes Section

After completing the algorithm tests for the Iris data, I returned to the 'Preprocess' tab, cleared the workspace, and loaded my second preinstalled dataset: diabetes.arff. I then repeated the classification process.

Naïve Bayes

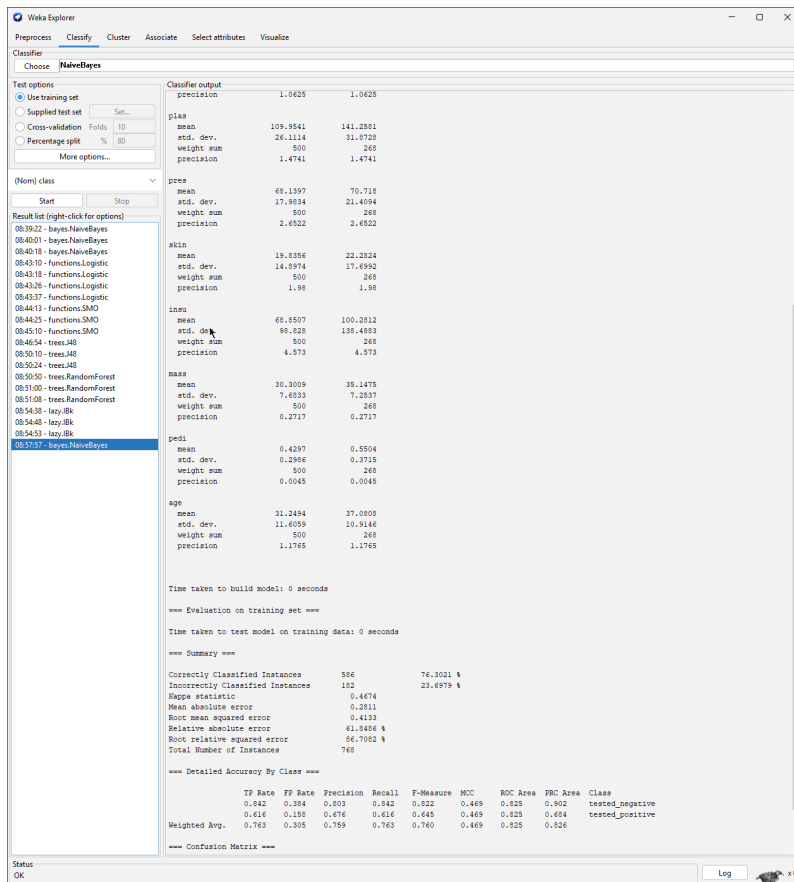


Figure 19: Naïve Bayes, Use Training Set

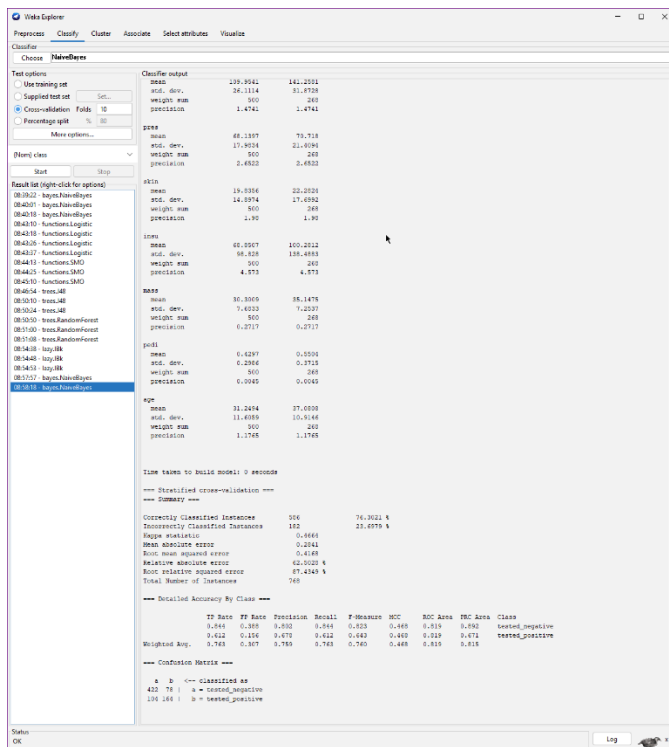


Figure 20: Naïve Bayes, Cross-validation (10 Folds)

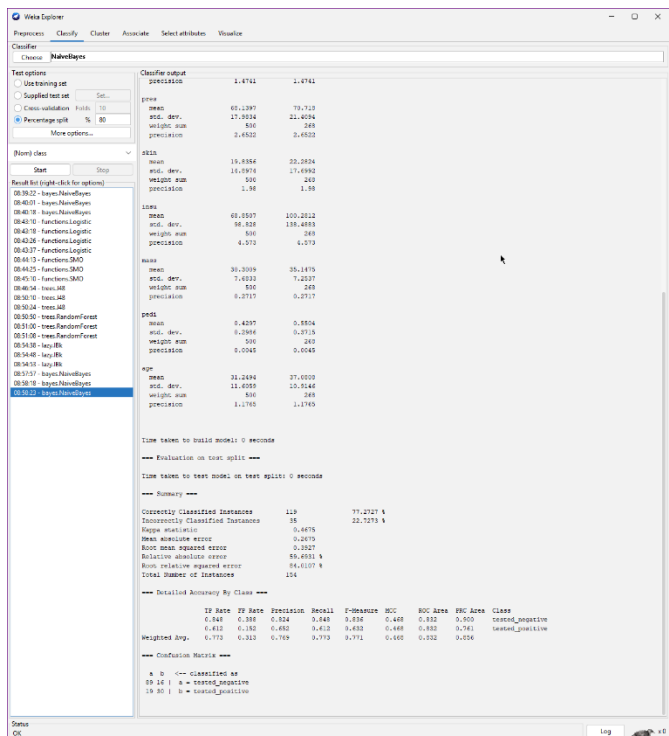


Figure 21: Naïve Bayes, Percentage Split (80%)

Logistic Regression

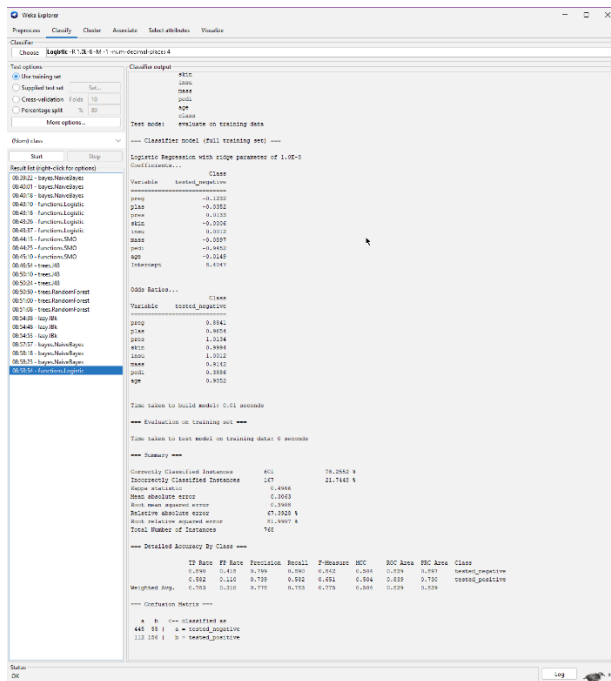


Figure 22: Logistic Regression, Use Training Set

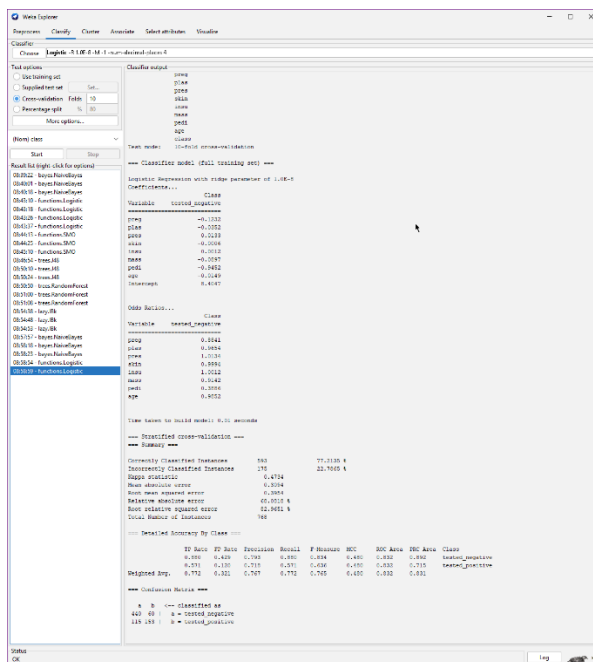


Figure 23: Logistic Regression, Cross-validation (10 Folds)

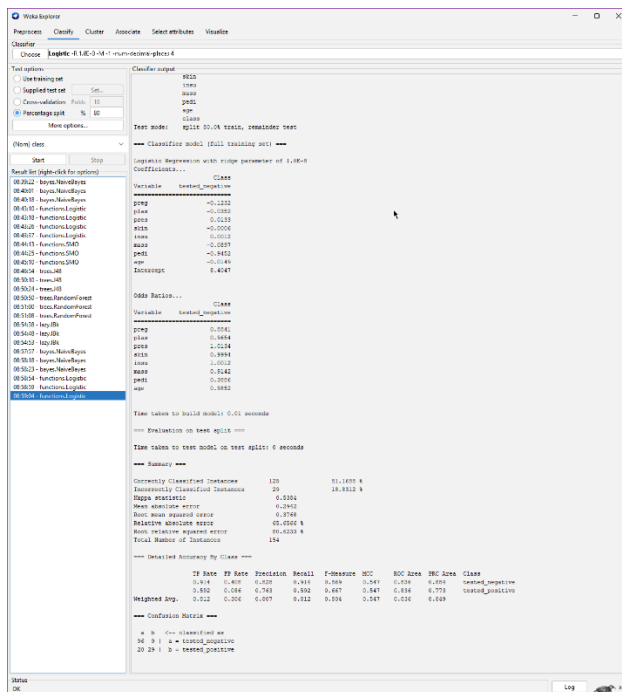


Figure 24: Logistic Regression, Percentage Split (80%)

SMO

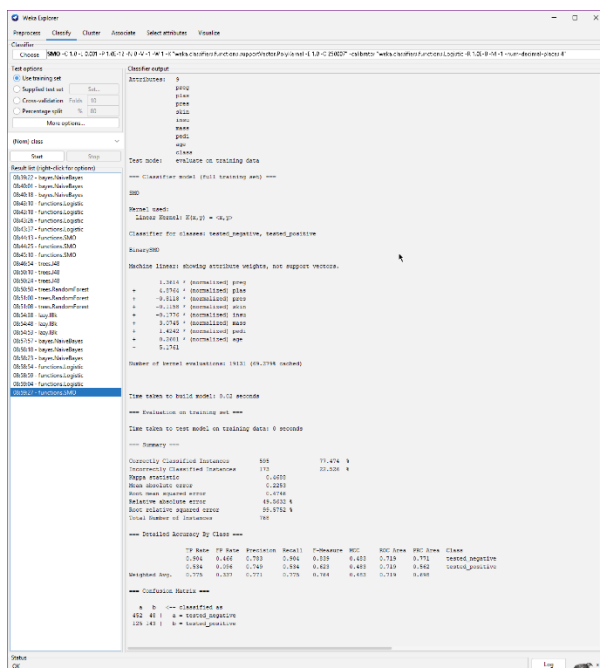
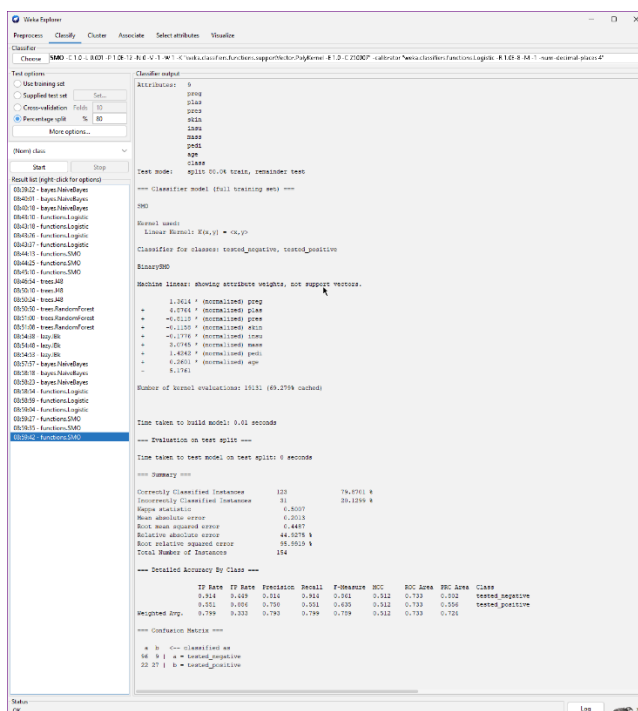
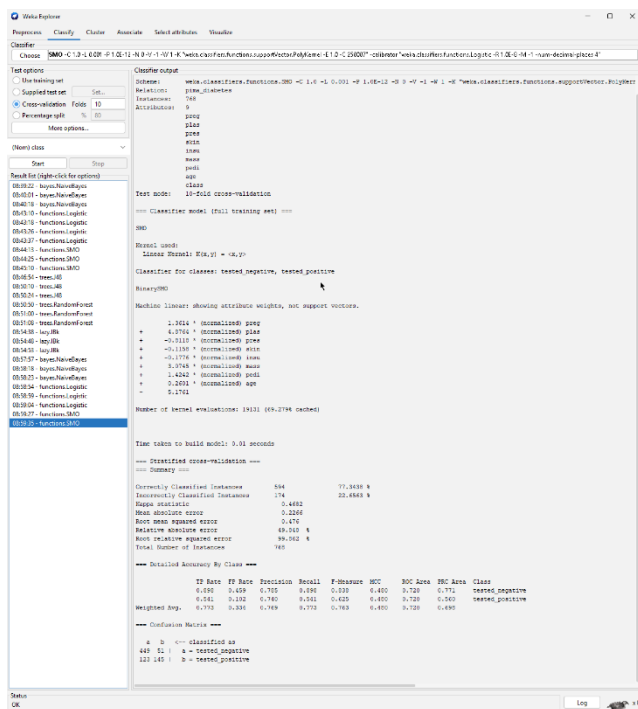


Figure 25: SMO, Use Training Set



K-Nearest Neighbor

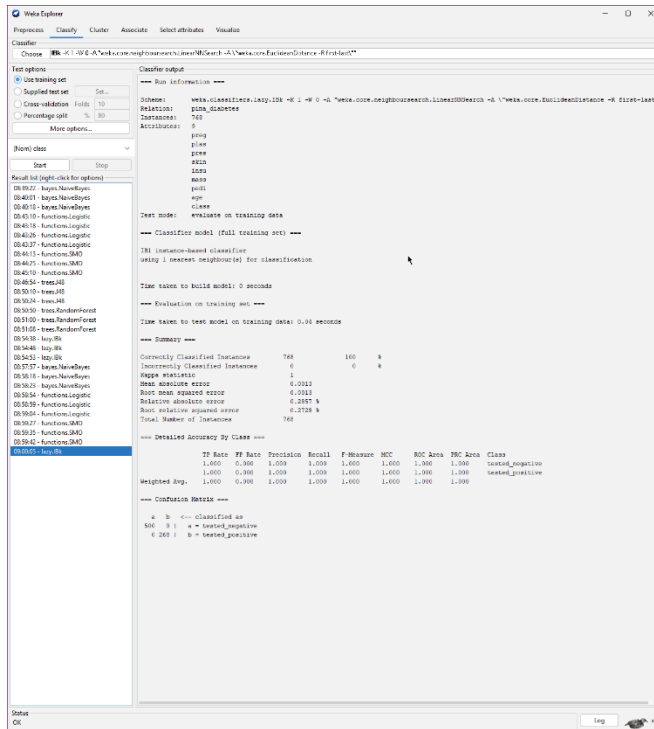


Figure 28: K-Nearest Neighbor, Use Training Set

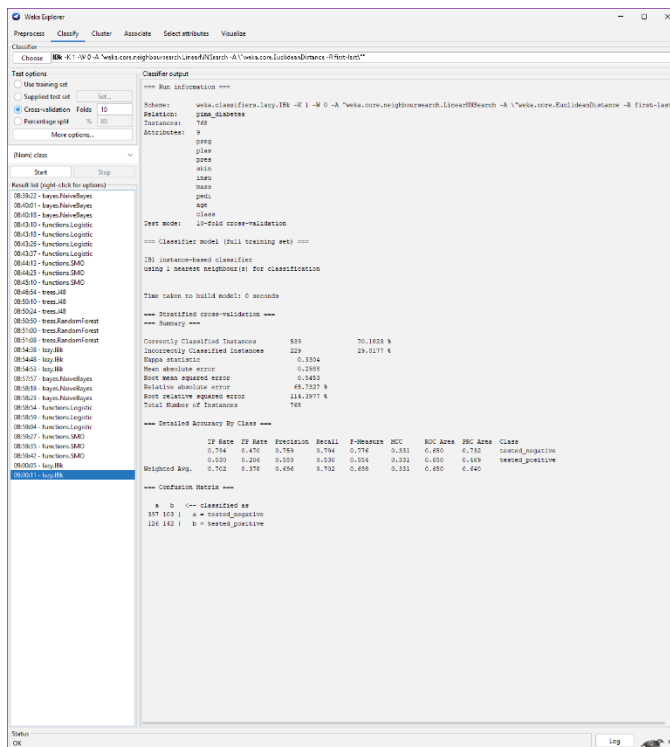


Figure 29: K-Nearest Neighbor, Cross-validation (10 Folds)

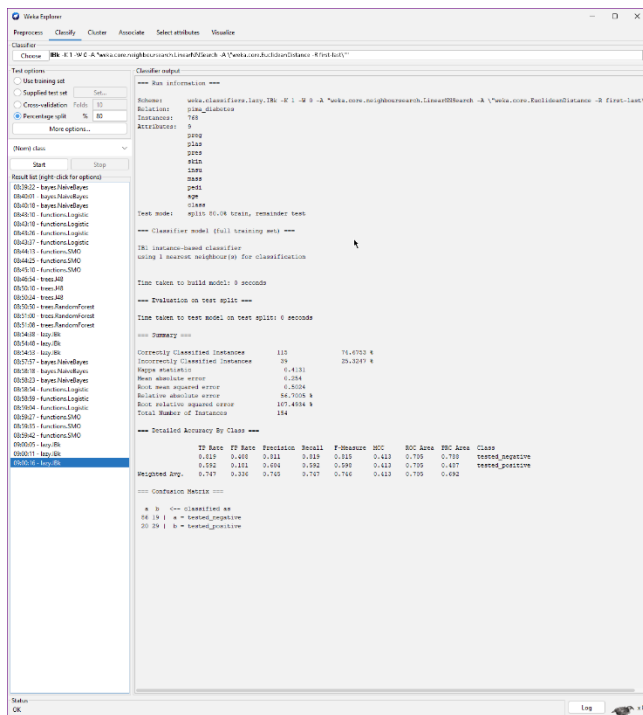


Figure 30: K-Nearest Neighbor, Percentage Split (80%)

J48

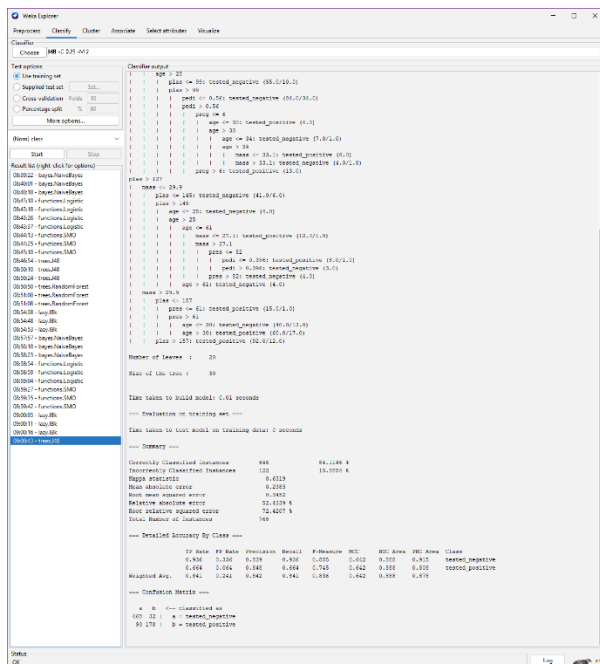


Figure 31: J48, Use Training Set

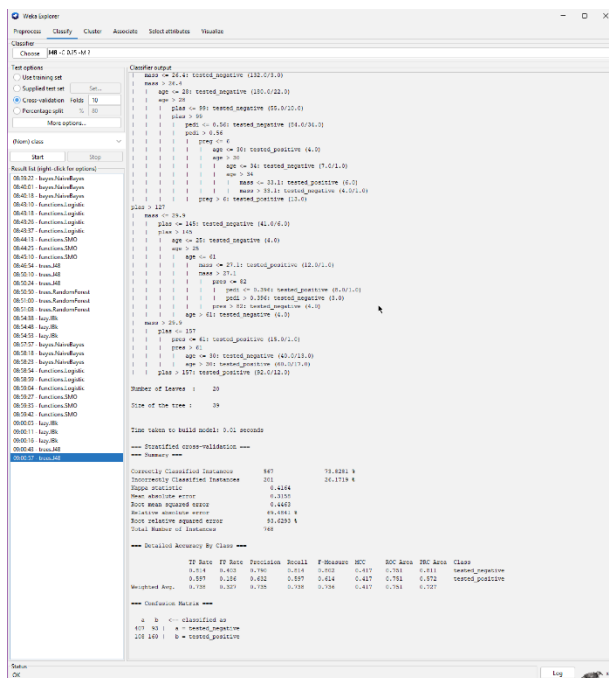


Figure 32: J48, Cross-validation (10 Folds)

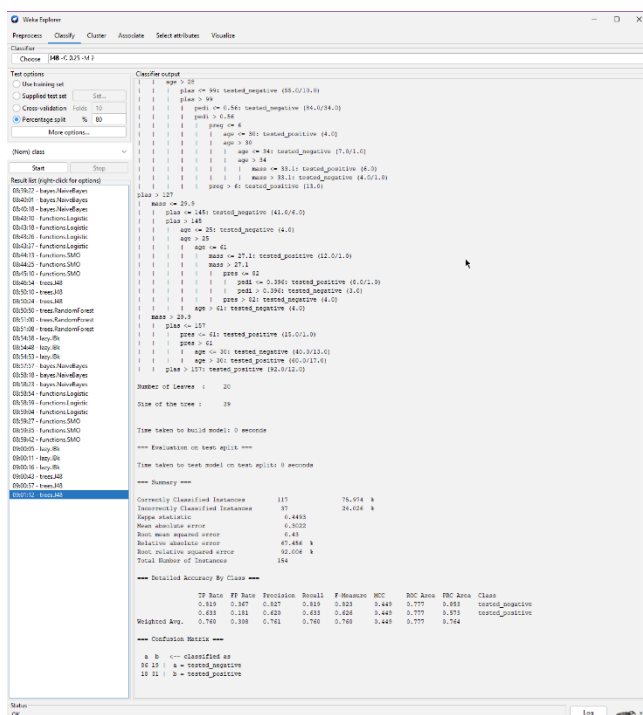


Figure 33: J48, Percentage Split (80%)

Random Forest

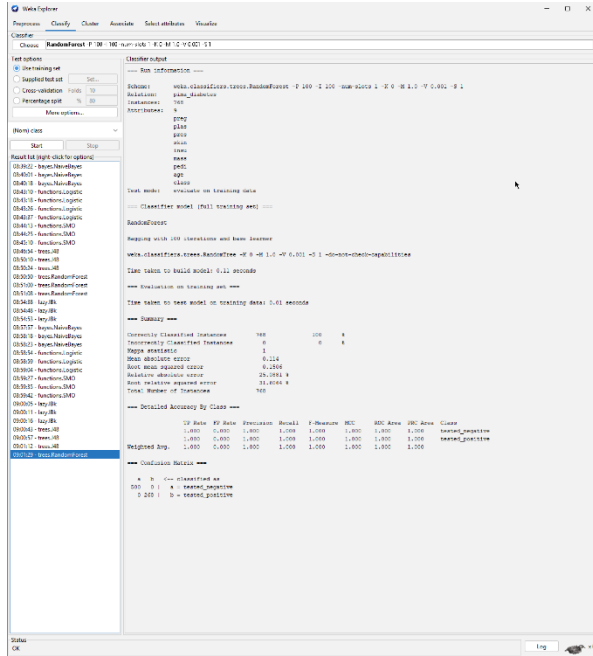


Figure 34: Random Forest, Use Training Set

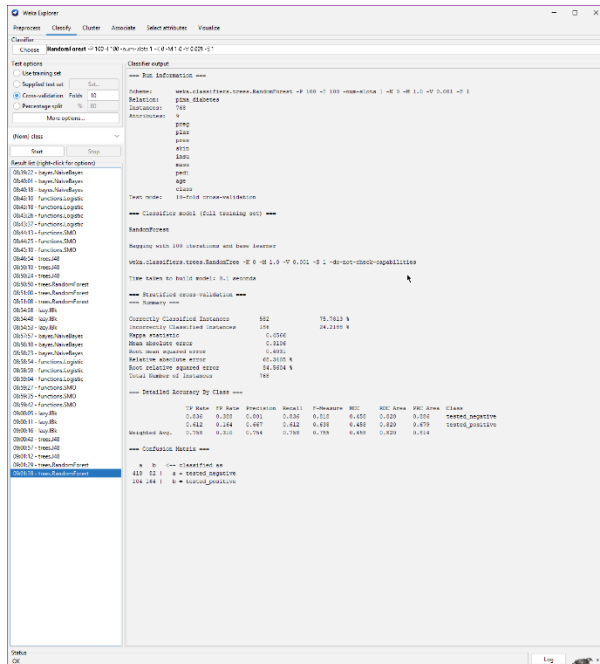


Figure 35: Random Forest, Cross-validation (10 Folds)

